

C4

Evals as a layer — and your first assignment

12 minutes

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Explain why evaluation is part of the stack, name the levels of eval, and preview A1 — a prompt + a 10-case eval harness (released Sun 16 Aug 2026, end of Week 2).

Why evals — because AI fails silently

No crash, no 404 — just a confident wrong answer.

- Traditional bug → crash / error you can catch.
- AI bug → a plausible answer that's simply wrong.
- You can't see it from the outside → you must measure it.
- Evals = your manufactured ground truth, run continuously.

Takeaway If you can't measure it, you can't ship it — evals are part of the stack.

Silent failure — three ways it hides

No crash, no 404 — just a confident wrong answer. Straight from your pre-reads.

Wrong, but fluent

Huyen · Reading 2

Unverifiable translation

A drifting fraud model

Why: no ground truth

Bias in the weights

Karpathy · Reading 3

Résumé model, quietly biased

Confident wrong label

Why: opaque weights

Confident hallucination

Husain · Reading 4

RAG invents a citation

Vibe-check passes

Why: plausible ≠ true

Takeaway Plausibility ≠ accuracy — the failure hides inside a correct-looking answer.

Three levels of evaluation

Start cheap and automatic; add judgement only where you need it. (LLM = Large Language Model · A/B = split-traffic test)

L1 · Assertions in code

deterministic true/false — schema-valid · exact-match · regex; run on every change

L2 · Human + LLM-as-judge

humans label traces; then an LLM 'judge' aligned to those labels (track agreement)

L3 · A/B testing (online)

split real traffic; compare guardrail metrics; mature product + big changes only

Takeaway L1 = automate what's checkable · L2 = judge what isn't · L3 = prove it in production.

L1 in practice — automate what's checkable

If you can write it as true/false in code, it's an L1 assertion. Everything else goes to a judge (L2).

Automate — L1 assertions

Deterministic; run on every change.

- Returns exactly 1 record; valid JSON to schema
- Required fields present; no null; no PII leak
- Cites ≥ 1 retrieved source; output $\leq N$ tokens
- Numeric answer within tolerance of expected

Judge instead — L2

No single right answer → human / LLM-judge.

- 'Is this answer helpful and correct?'
- Tone, style, faithfulness to the context
- Is the summary complete and on-topic?
- Anything needing human-like judgement

Takeaway A1 = a 10-case harness: L1 assertions like the left, PLUS at least one L2 judge check (next slide).

LLM-as-judge & eval metrics

FOR BUILDERS

For the builders (skim for a non-technical room): how to score quality when there's no single right answer.

LLM-as-judge, done right

An LLM scores outputs against a rubric.

- Pairwise (A vs B) beats pointwise scoring
- Rubric + few-shot; ask for reasons, then a score
- ALIGN to humans first; track agreement (Cohen's κ)
- Watch bias: position · verbosity · self-preference

Metrics that matter

Pick per task; automate the checkable ones.

- Deterministic: exact-match · F1 · schema-valid · pass@k
- Retrieval: recall@k · precision@k · MRR · nDCG (C3)
- Generation: faithfulness/groundedness · answer-relevance
- Tools: RAGAS · promptfoo · LangSmith · DeepEval

Takeaway Automate what's checkable (L1); use an aligned LLM-judge for the rest (L2) — and prove the judge agrees with humans.

Tie the metric to the business

A model metric nobody cares about won't survive contact with a budget.

eval metric (e.g. recall@k, answer accuracy) → product outcome (fewer escalations) →
business metric (\$ cost-to-serve, revenue, risk)

Takeaway Every eval should ladder up to a number the business already tracks.

Evals across industries

Same discipline, life-or-death stakes vary.

Healthcare

illustrative

Risk: hallucinated drug dose

Eval: clinical-accuracy checks

Human-in-the-loop; cost catastrophic

Telecom

illustrative

Risk: churn / anomaly model drift

Eval: drift detection

Metric: precision@alert

Banking

DBS

Risk: AML miss / bad answer

Eval: P-U-R-E 'explainable'

Metric: fraud \$ stopped

Takeaway The stakes set the eval bar — a chatbot and a dosing tool are not the same risk.

A1 — Foundations Build (released Sun 16 Aug 2026)

Individual · code + short write-up · due Sun 30 Aug 2026, 11:59 PM · Rubric 1.

- (i) AI-type selection — scikit-learn classifier vs a foundation-model API (from C1).
- (ii) A minimal RAG build over a small corpus (today's Colab).
- (iii) A prompt + a 10-case eval harness — L1 assertions plus at least one L2 (LLM-judge / human) check — for a task in your industry.
- Submit: code + short write-up + a brief self-appraisal. Marked on Rubric 1.

Takeaway A1 = the three things you did in Class 1–2, on a problem you choose.

QUICK CHECK 1 • WOOC LAP

Which of these is a usable L1 (level-1) eval assertion?

A) "Is the answer good?"

B) "Feels helpful"

C) "Returns exactly 1 record"

D) "Sounds confident"

Vote in Wooclap — answer on the next slide.

QUICK CHECK 1 · ANSWER

C — "Returns exactly 1 record".

Mechanically checkable true/false = L1. The rest are vibes (an L2 judgement, not an L1 test).

QUICK CHECK 2 • WOOCCLAP

Your LLM-as-judge agrees with human labels only ~60% of the time. What do you do FIRST?

A) Trust the judge — it scales

B) Switch to a bigger judge model

C) Drop evals; ship and watch

D) Align it: sharpen the rubric + add human-labelled examples, re-check agreement

Vote in Wooclap — answer on the next slide.

QUICK CHECK 2 · ANSWER

D — align the judge to humans first.

Sharpen the rubric + add human-labelled examples, then re-check agreement (κ). An unaligned judge just launders bias.

QUICK CHECK 3 • WOOC LAP

Retrieval recall@k is high, but answers are still wrong. Which eval catches it?

A) Faithfulness / groundedness — is the answer supported by the chunks?

B) recall@k again

C) Latency

D) A bigger embedding model

Vote in Wooclap — answer on the next slide.

QUICK CHECK 3 · ANSWER

A — faithfulness / groundedness.

recall@k judges retrieval; when that's fine but answers are wrong, the generation is unfaithful to the context.

QUICK CHECK 4 • WOOC LAP

Answers are faithful to the context but users say they're off-topic. Which metric were you missing?

A) recall@k

B) Token cost

C) Answer-relevance — does it actually address the question?

D) Exact-match

Vote in Wooclap — answer on the next slide.

QUICK CHECK 4 · ANSWER

C — answer-relevance.

Faithful $\neq$ on-topic: an answer can be supported by the context yet not address the question. Measure both.

C4 · TAKEAWAY

**Evaluation is a layer of the stack, not an afterthought —
you ship what you can measure.**

That closes the stack: compute → ... → evals. A1 releases Sun 16 Aug 2026; project statement due Sun 23 Aug 2026.

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Coming up — Class 3

[NEXT](#)

Open the model, then steer it with prompts you can measure — a level up in depth from today (details firm up nearer the date).

- 1 **Inside the model** — tokens, attention, transformers, scaling laws — why capability emerges with scale.
- 2 **Prompting craft** — zero/few-shot, chain-of-thought, self-consistency, structured output.
- 3 **Eval-driven iteration** — build an eval set and improve the prompt against it — measure, don't guess.
- 4 **Token economics** — what a prompt really costs, and how to cut it.
- 5 **Hands-on (indicative)** — a step up from today — multi-step, structured-output prompting, iterated against your own evals.
- 6 **Milestone** — project problem statement — submit by Sun 23 Aug 2026, 11:59 PM.

Throughline: Prompting is programming in words — and evals are how you know it works.